

ORIGINAL RESEARCH

Bivalirudin Versus Heparin in Low and Non-Low Bleeding Risk Patients Undergoing Primary PCI for STEMI

The BRIGHT-4 Trial

Miaohan Qiu, MD,^{a,*} Yixuan Duan, MD,^{a,b,*} Xiaojian Feng, MD,^c Kui Chen, MD,^d Dingbao Zhang, MD,^e Yuncheng Zou, MD,^f Xiaohui Zheng, MD,^g Gang Zhao, MD,^h Nailiang Tian, MD,ⁱ Zeqi Zheng, MD,^j Xiaoping Peng, MD,^j Qing Yang, MD,^k Zhenyang Liang, MD,^a Yi Li, MD,^a Yaling Han, MD,^a Gregg W. Stone, MD^l

ABSTRACT

BACKGROUND In the BRIGHT-4 trial, among 6,016 patients with ST-segment elevation myocardial infarction (STEMI) undergoing primary percutaneous coronary intervention (PCI) with a radial artery approach, procedural anticoagulation with bivalirudin plus a post-PCI high-dose infusion for 2 to 4 hours reduced the 30-day primary composite outcome of all-cause death or Bleeding Academic Research Consortium (BARC) types 3 to 5 bleeding, as well as death and bleeding individually, compared with heparin monotherapy.

OBJECTIVES We sought to determine whether the benefits of bivalirudin apply principally to patients who are at low bleeding risk (LBR) as well as non-LBR.

METHODS In a prespecified analysis from BRIGHT-4, outcomes were examined by baseline bleeding risk, with LBR defined as a CRUSADE score <30.

RESULTS At baseline, 4,581 patients (76.1%) were categorized as LBR. Non-LBR patients had higher rates of the 30-day primary endpoint (8.6% vs 2.2%; HR: 4.08 [95% CI: 3.14-5.31]; $P < 0.0001$), driven by both greater mortality and BARC types 3 to 5 bleeding. In non-LBR patients, the primary outcome occurred in 8.1% of patients randomized to bivalirudin vs 9.2% of those randomized to heparin (difference: -1.1% [95% CI: -4.0% to 1.8%]; HR: 0.88 [95% CI: 0.62-1.26]). In LBR patients, the primary outcome occurred in 1.4% of patients randomized to bivalirudin vs 2.9% of those randomized to heparin (difference: -1.5% [95% CI: -2.3% to -0.6%]; HR: 0.49 [95% CI: 0.32-0.75]) ($P_{\text{absolute interaction}} = 0.81$; $P_{\text{relative interaction}} = 0.04$). The effects of bivalirudin compared with heparin in reducing all-cause death were as robust in LBR patients compared with non-LBR patients ($P_{\text{absolute interaction}} = 0.67$; $P_{\text{relative interaction}} = 0.06$).

CONCLUSIONS Among patients with STEMI undergoing primary PCI with radial artery access, procedural anticoagulation with bivalirudin plus a high-dose post-PCI infusion for 2 to 4 hours reduced the 30-day risk of all-cause death and major bleeding in patients at low bleeding risk as well as in patients at higher-risk of bleeding. (Bivalirudin With Prolonged Full Dose Infusion Versus Heparin Alone During Emergency PCI [BRIGHT-4; NCT03822975]) (JACC. 2026;■:■-■) © 2026 by the American College of Cardiology Foundation.

From the ^aState Key Laboratory of Frigid Zone Cardiovascular Disease, Department of Cardiology, General Hospital of Northern Theater Command, Shenyang, Liaoning, China; ^bAir Force Medical University, Xi'an, Shaanxi, China; ^cDepartment of Cardiology, Yueyang Central Hospital, Yueyang, Hunan, China; ^dDepartment of Cardiology, First Affiliated Hospital of Zhengzhou University, Zhengzhou, Henan, China; ^eDepartment of Cardiology, People's Hospital of Pingchang County, Pingchang, Sichuan, China; ^fDepartment of Cardiology, Yunnan St John's Hospital, Kunming, Yunnan, China; ^gDepartment of Emergency, Fuwai Central China Cardiovascular Hospital, Central China Fuwai Hospital of Zhengzhou University, Henan Provincial People's Hospital, Zhengzhou, Henan, China; ^hDepartment of Cardiology, First Hospital of Liangshan, Xichang, Sichuan, China; ⁱDepartment of

ABBREVIATIONS AND ACRONYMS

LBR = low bleeding risk

MACCE = major adverse cardiac or cerebral events

NACE = net adverse clinical events

PCI = percutaneous coronary intervention

STEMI = ST-segment elevation myocardial infarction

TVR = target vessel revascularization

Patients with ST-segment elevation myocardial infarction (STEMI) undergoing primary percutaneous coronary intervention (PCI) are at risk for early recurrent ischemic and major bleeding events, both of which affect early and late survival.^{1,2} The choice of access site, antiplatelet therapy selection, and type of procedural anticoagulation during primary PCI, among other factors, have a major influence on both recurrent ischemic and bleeding events. Based on previous studies,³⁻⁸ heparin monotherapy and bivalirudin with a high-dose post-PCI infusion for 2 to 4 hours emerged as the optimal regimen to balance intra- and postprocedural thrombotic and bleeding complications after primary PCI in STEMI.⁹ In the BRIGHT-4 (Bivalirudin With Prolonged Full Dose Infusion Versus Heparin Alone During Emergency PCI; [NCT03822975](#)) trial, 6,016 STEMI patients undergoing primary PCI with radial artery access were randomized to procedural anticoagulation with bivalirudin plus a post-PCI high-dose infusion for 2 to 4 hours vs heparin monotherapy.¹⁰ Bivalirudin reduced the 30-day primary composite outcome of all-cause death or Bleeding Academic Research Consortium (BARC) types 3 to 5 bleeding, as well as death and bleeding individually and stent thrombosis, compared with heparin anticoagulation. On the basis of these and earlier data,⁹ procedural anticoagulation with bivalirudin plus a 2 to 4-hour post-PCI high-dose infusion was afforded a class I recommendation as an alternative to unfractionated heparin to reduce mortality and bleeding during primary PCI in STEMI in the most recent 2025 U.S. societal guidelines.¹¹

Patients with STEMI with features predisposing to bleeding are at increased risk not only for bleeding but also for reinfarction and death after primary PCI.^{12,13} Whether the benefits of bivalirudin in STEMI extend to patients at low bleeding risk (LBR), especially those undergoing PCI with radial artery access (a strategy that also reduces access site-related bleeding and mortality in STEMI¹¹) is unknown. We therefore performed a prespecified substudy from the BRIGHT-4 trial (in which radial artery access was

used in 93.1% of patients) to examine the 30-day outcomes of bivalirudin plus a high-dose post-PCI infusion for 2 to 4 hours compared with heparin monotherapy in patients with STEMI undergoing primary PCI, according to their bleeding risk.

METHODS

The trial was approved by the Institutional Review Board or ethics committee at each participating center. The study was performed in accordance with the principles of the Declaration of Helsinki. Written informed consent was provided by all patients or their legal representatives before random assignment.

STUDY POPULATION. The BRIGHT-4 trial design and principal outcomes have been previously published.^{10,14} In brief, BRIGHT-4 was an investigator-initiated, open-label, active drug-controlled randomized trial conducted at 87 clinical centers in 63 cities in China. Patients of any age presenting with STEMI within 48 hours of symptom onset undergoing primary PCI were randomly assigned to receive bivalirudin with a prolonged high-dose infusion for 2 to 4 hours after the procedure or unfractionated heparin alone in a 1:1 ratio according to an interactive web response system.

TREATMENTS. Study medications were administered in accordance with the randomized assignment before coronary angiography in the catheterization laboratory. Patients in the bivalirudin group received a 0.75 mg/kg intravenous bolus followed by a continuous infusion of 1.75 mg/kg per hour during PCI (adjusted for renal insufficiency) and for 2 to 4 hours afterward. Those allocated to heparin were treated with an initial bolus of 70 U/kg. The activated clotting time was measured 5 minutes after the first bolus of each agent and additional bivalirudin (0.30 mg/kg) or heparin (1,000 U) was administered if the activated clotting time was <225 seconds. In both groups, use of a glycoprotein IIb/IIIa inhibitor was allowed only in case of development of refractory thrombotic complications during the procedure.

All patients received dual antiplatelet therapy with aspirin plus either clopidogrel or ticagrelor. Other

Cardiology, Nanjing First Hospital, Nanjing Medical University, Nanjing, Jiangsu, China; ¹Department of Cardiology, First Affiliated Hospital of Nanchang University, Nanchang, Jiangxi, China; ²Department of Cardiology, Tianjin Medical University General Hospital, Tianjin, China; and the ³Icahn School of Medicine at Mount Sinai, New York, New York, USA. *These authors contributed equally to this work.

The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the [Author Center](#).

Manuscript received March 3, 2026; revised manuscript received May 11, 2026, accepted June 15, 2026.

pharmacologic treatments were administered according to guideline-directed management. The radial artery was the preferred access route, and primary PCI was performed using standard techniques.

BLEEDING RISK. For the prespecified analysis, bleeding risk in BRIGHT-4 was assessed at hospital admission with the use of the CRUSADE (Can Rapid Risk Stratification of Unstable Angina Patients Suppress Adverse Outcomes With Early Implementation of the ACC/AHA Guidelines) bleeding risk score, which is calculated from 8 baseline clinical and laboratory variables available, including sex, diabetes, history of peripheral arterial disease, congestive heart failure at admission, heart rate, systolic blood pressure, hematocrit, and Cockcroft-Gault-estimated creatinine clearance.¹⁵ The CRUSADE score ranges from 1 to 96 points, with <30 points signifying patients at LBR and ≥30 points identifying patients at moderate or high bleeding risk (non-LBR). Numerous studies have demonstrated that this cutoff identifies patients with STEMI at greatly increased risk of bleeding and death.¹⁶⁻¹⁸ Sensitivity analyses were also performed using the PRECISE-DAPT score (<25 and ≥25 points signifying LBR and non-LBR patients, respectively) and the ARC-HBR criteria (≥1 major ARC-HBR criterion or ≥2 minor ARC-HBR criteria signifying non-LBR).^{19,20}

OUTCOMES. The trial endpoints and definitions were identical to those of the BRIGHT-4 trial.¹⁰ The primary outcome was the 30-day composite of all-cause death or BARC types 3 to 5 bleeding. Secondary outcomes were the individual components of the primary endpoint, major adverse cardiac or cerebral events (MACCE; the composite of all-cause death, recurrent myocardial infarction, ischemia-driven target vessel revascularization [TVR], or stroke) and its components, stent thrombosis (defined by Academic Research Consortium definite or probable criteria),²¹ BARC types 2 to 5 bleeding, the composite of all-cause death or BARC types 2 to 5 bleeding, acquired thrombocytopenia, and net adverse clinical events (NACE; the composite of MACCE or BARC types 3-5 bleeding).^{22,23} All primary and secondary events were adjudicated by an independent clinical events committee masked to the therapy assignment.

STATISTICAL ANALYSIS. With 6,000 randomized patients, BRIGHT-4 was powered to demonstrate a reduction in the 30-day composite of all-cause death or major bleeding with bivalirudin compared with heparin.¹⁰ For the present analysis, interaction testing was performed to determine whether the CRUSADE and other bleeding risk scores affected the

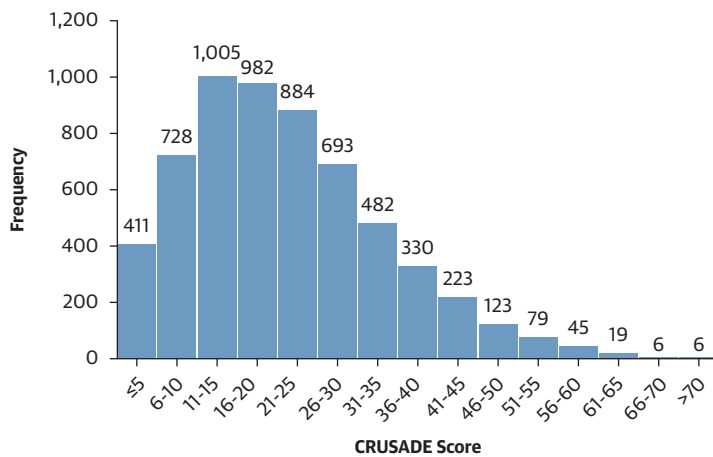
absolute and relative outcomes of randomization to bivalirudin vs heparin. The interactions were represented graphically showing the HR and absolute difference for the treatment effect of bivalirudin vs heparin across the range of bleeding risk score. Receiver operating characteristic curves were constructed to evaluate the performance of the CRUSADE score in predicting outcomes, and the area under the curve was calculated to quantify discriminative ability.

Continuous variables are presented as the mean ± SD or the median with IQR and were compared by means of a Student's *t*-test or the Wilcoxon rank-sum test as appropriate. Categorical variables are presented as n (%) and were compared by means of the chi-square test or Fisher exact test. Time-to-first event rates were estimated using Kaplan-Meier methodology and were compared by means of the log-rank test. HRs and 95% CIs were established from a Cox model. A *P* value of <0.05 was considered to be statistically significant, and all *P* values are 2 sided. All statistical analyses were performed with SAS version 9.4 (SAS Institute, Inc).

RESULTS

BASELINE CHARACTERISTICS. CRUSADE scores were determined for all 6,016 patients, among whom the median score was 20 (Q1-Q3: 12-29). As shown in the histogram in [Figure 1](#), using standard cutoffs, 3,126 patients (52.0%) were at very low risk for bleeding (≤20 points), 1,577 (26.2%) were at low risk (21-30 points), 812 (13.5%) were at moderate risk (31-40 points), 346 (5.8%) were at high risk (41-50 points), and 155 (2.6%) were at very high risk (>50 points). With the prespecified cutoff, 1,435 patients (23.9%) were non-LBR (CRUSADE score ≥30), 728 of whom were randomized to bivalirudin and 707 to heparin, and 4,581 patients (76.1%) were LBR (CRUSADE score <30), 2,281 of whom were randomized to bivalirudin and 2,300 to heparin ([Figure 2](#)).

As presented in [Supplemental Table 1](#), non-LBR patients were older, more frequently male, had higher rates of hypertension, diabetes, previous myocardial infarction, anemia, and chronic kidney disease, and presented with worse Killip class, with longer symptom onset-to-first medical contact and tertiary hospital arrival times. As presented in [Table 1](#), baseline characteristics were generally well balanced between the randomized treatment arms in both CRUSADE strata. Among non-LBR patients, however, anemia was slightly more common in the heparin group compared with the bivalirudin group, although the mean hemoglobin levels were similar.

FIGURE 1 Histogram of CRUSADE Scores Among 6,016 Enrolled Patients

STUDY MEDICATIONS AND PROCEDURES. As presented in [Supplemental Table 2](#), radial artery access was slightly less common in non-LBR than in LBR patients (89.8% vs 94.1%; $P < 0.0001$). Non-LBR patients received lower doses of heparin but a slightly longer post-PCI bivalirudin infusion compared with LBR patients. Clopidogrel rather than ticagrelor was more frequently used in non-LBR than in LBR patients (although most patients in both groups were treated with ticagrelor). Tirofiban was used for procedural thrombotic complications slightly more frequently in non-LBR patients.

As presented in [Table 2](#), study medications and procedures were generally well matched between treatment arms. The median peak activated clotting

time was greater in patients administered bivalirudin regardless of bleeding risk, and additional boluses of study medications were more frequently administered in patients treated with heparin. Bail-out tirofiban use was more frequent in the LBR group in heparin-treated compared with bivalirudin-treated patients. In non-LBR patients, post-PCI Thrombolysis in Myocardial Infarction flow grade 3 was achieved more frequently with bivalirudin than with heparin.

CLINICAL OUTCOMES ACCORDING TO BLEEDING RISK.

The CRUSADE score demonstrated good discrimination for predicting the composite endpoint, all-cause death and major bleeding ([Supplemental Figure 1](#)). BARC types 3 to 5 bleeding occurred in 0.9% and 0.3% of non-LBR and LBR patients, respectively (HR: 2.60 [95% CI: 1.25-5.40]; $P = 0.01$) ([Table 3](#)). The increase in major bleeding in non-LBR patients was due to greater non-access site-related bleeding; access site-related bleeding was infrequent in both non-LBR and LBR patients. BARC types 2 to 5 bleeding also was increased in non-LBR patients. Notably, non-LBR patients were at markedly increased risk for 30-day all-cause death (8.1% vs 2.0%; HR: 4.19 [95% CI: 3.18-5.51]; $P < 0.0001$). The 30-day primary composite endpoint of death or BARC types 3 to 5 bleeding was therefore increased in non-LBR compared with LBR patients (8.6% vs 2.2%; HR: 4.08 [95% CI: 3.14-5.31]; $P < 0.0001$). Non-LBR patients were also at higher risk for cardiovascular death, reinfarction and stent thrombosis, MACCE and NACE, but not TVR, stroke, or thrombocytopenia ([Table 3](#)).

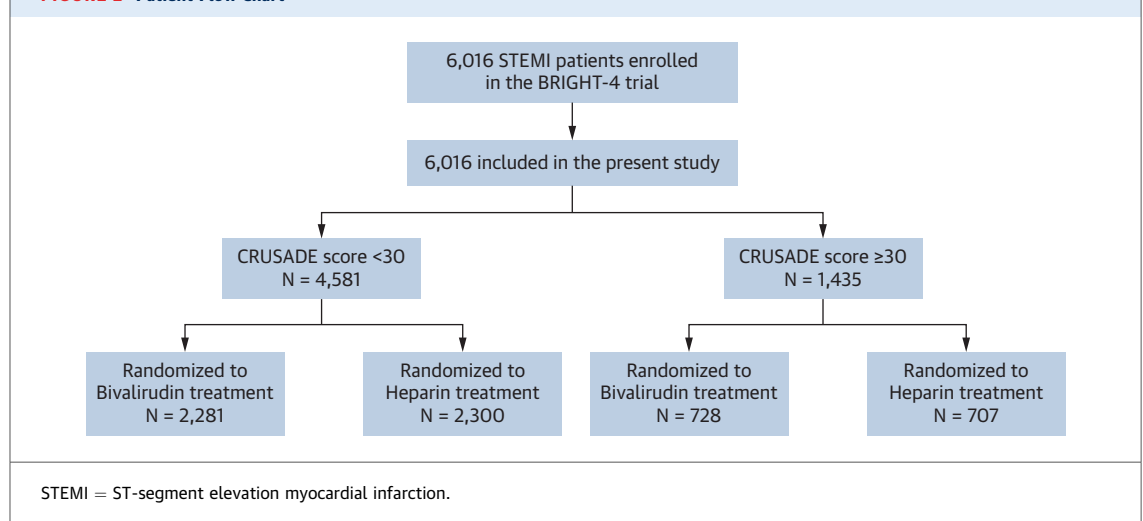
FIGURE 2 Patient Flow Chart

TABLE 1 Baseline Characteristics According to the CRUSADE Score and Randomized Anticoagulation Type

	CRUSADE Score <30 (n = 4,581)			CRUSADE Score ≥30 (n = 1,435)		
	Heparin (n = 2,300)	Bivalirudin (n = 2,281)	SMD (95% CI)	Heparin (n = 707)	Bivalirudin (n = 728)	SMD (95% CI)
Age, y	58.7 ± 11.9	58.6 ± 11.7	−0.01 (−0.07 to 0.05)	66.6 ± 11.2	66.4 ± 11.4	−0.02 (−0.12 to 0.09)
Male	2,009 (87.4)	1,994 (87.4)	0.00 (−0.06 to 0.06)	363 (51.3)	356 (48.9)	−0.05 (−0.05 to 0.15)
Body mass index, kg/m ²	25.2 ± 3.8	24.8 ± 3.6	−0.09 (−0.15 to −0.03)	24.3 ± 3.7	24.5 ± 3.5	0.05 (−0.05 to 0.15)
Medical history						
Hypertension	1,078 (46.9)	1,093 (47.9)	0.02 (−0.04 to 0.08)	440 (62.2)	471 (64.7)	0.05 (−0.05 to 0.15)
Diabetes mellitus	411 (17.9)	387 (17.0)	−0.02 (−0.08 to 0.03)	287 (40.6)	280 (38.5)	−0.04 (−0.15 to 0.06)
Smoking			0.04 (−0.01 to 0.10)			0.15 (0.04 to 0.25)
Active	1,170 (50.9)	1,178 (51.6)		222 (31.4)	182 (25.0)	
Former	201 (8.7)	172 (7.5)		52 (7.4)	65 (8.9)	
Never	929 (40.4)	931 (40.8)		433 (61.2)	481 (66.1)	
Previous MI	147 (6.4)	130 (5.7)	−0.03 (−0.09 to 0.03)	50 (7.1)	58 (8.0)	0.03 (−0.07 to 0.14)
Previous PCI	148 (6.4)	127 (5.6)	−0.04 (−0.09 to 0.02)	38 (5.4)	59 (8.1)	0.11 (0.01 to 0.21)
Previous stroke	198 (8.6)	198 (8.7)	0.00 (−0.06 to 0.06)	146 (20.7)	155 (21.3)	0.02 (−0.09 to 0.12)
Killip class ^a			0.05 (−0.01 to 0.11)			0.08 (−0.03 to 0.18)
I	1,552 (67.5)	1,525 (66.9)		305 (43.1)	293 (40.3)	
II	562 (24.4)	596 (26.1)		266 (37.6)	288 (39.6)	
III	160 (7.0)	138 (6.1)		67 (9.5)	81 (11.1)	
IV	26 (1.1)	22 (1.0)		69 (9.8)	66 (9.1)	
Hemoglobin, g/dL	14.3 ± 1.7	14.3 ± 1.7	−0.03 (−0.09 to 0.03)	13.1 ± 1.8	13.2 ± 1.8	0.08 (−0.02 to 0.19)
Anemia ^b	388 (16.9)	392 (17.2)	0.01 (−0.05 to 0.07)	238 (33.7)	207 (28.4)	−0.11 (−0.22 to −0.01)
Platelet count, 10 ⁹ /L	227.5 ± 65.1	230.5 ± 72.0	0.04 (−0.01 to 0.10)	228.6 ± 68.8	225.6 ± 74.1	−0.04 (−0.15 to 0.06)
eGFR, mL/min/1.73 m ^{2c}	115.0 ± 29.7	115.4 ± 30.1	0.01 (−0.04 to 0.07)	74.1 ± 24.0	72.6 ± 23.2	−0.06 (−0.17 to 0.04)
eGFR <60 mL/min/1.73 m ²	7 (0.3)	2 (0.1)	−0.05 (−0.11 to 0.01)	198 (28.0)	209 (28.7)	0.02 (−0.09 to 0.12)
Symptom onset to first medical contact, h	2.1 (1.0-4.5)	2.0 (1.0-4.5)	−0.02 (−0.09 to 0.04)	2.5 (1.0-5.4)	2.4 (1.1-4.9)	−0.09 (−0.21 to 0.03)
Patients transferred from a nontertiary hospital	797 (35.1)	831 (36.7)	0.03 (−0.02 to 0.09)	240 (34.7)	253 (35.7)	0.02 (−0.08 to 0.13)
Symptom onset to tertiary hospital arrival, h	3.2 (1.7-6.5)	3.2 (1.7-6.4)	−0.02 (−0.08 to 0.05)	3.7 (1.8-7.2)	3.4 (1.8-6.4)	−0.08 (−0.19 to 0.04)
CRUSADE risk score	16.2 ± 7.4	16.1 ± 7.4	−0.01 (−0.07 to 0.05)	38.9 ± 8.0	39.5 ± 8.4	0.07 (−0.04 to 0.17)
PRECISE-DAPT score	12 (7-18)	12 (8-18)	0.03 (−0.03 to 0.09)	24 (17-32)	24 (17-32)	−0.02 (−0.12 to 0.08)
≥25	236 (10.3)	254 (11.1)	0.03 (−0.03 to 0.09)	342 (48.4)	356 (48.9)	0.01 (−0.09 to 0.11)
ARC-HBR, non-LBR	185 (8.0)	222 (9.7)	0.06 (0.00 to 0.12)	237 (33.5)	232 (31.9)	−0.04 (−0.14 to 0.07)

Values are mean ± SD, n (%), or median (Q1-Q3), unless otherwise indicated. ^aDefined as follows: class I, no clinical signs of heart failure; class II, rales or crackles in the lungs, a third heart sound, or an elevated jugular venous pressure; class III, pulmonary edema; and class IV, cardiogenic shock or hypotension with evidence of peripheral vasoconstriction. ^bDefined as a hemoglobin <13 g/dL in men and <12 g/dL in women. ^cCalculated by the formula: $186 \times (\text{serum creatinine [mg/dL]})^{-1.154} \times (\text{age})^{-0.203} \times (0.742 \text{ if female})$.

ARC-HBR = Academic Research Consortium—High Bleeding Risk; CRUSADE = Can Rapid Risk Stratification of Unstable Angina Patients Suppress Adverse Outcomes With Early Implementation of the ACC/AHA Guidelines; eGFR = estimated glomerular filtration rate; LBR = low bleeding risk; MI = myocardial infarction; PCI = percutaneous coronary intervention; PRECISE-DAPT = Predicting Bleeding Complications in Patients Undergoing Stent Implantation and Subsequent Dual Antiplatelet Therapy; SMD = standardized mean difference.

CLINICAL OUTCOMES ACCORDING TO ANTICOAGULATION RANDOMIZATION.

As presented in [Table 4](#) and [Figure 3](#), in LBR patients, the primary outcome occurred in 33 (1.4%) treated with bivalirudin vs 67 (2.9%) treated with heparin (difference: −1.5% [95% CI −2.3% to −0.6%]; HR: 0.49 [95% CI: 0.32-0.75]). In non-LBR patients, the 30-day primary outcome occurred in 59 (8.1%) treated with bivalirudin vs 65 (9.2%) treated with heparin (difference: −1.1% [95% CI: −4.0% to 1.8%]; HR: 0.88 [95% CI: 0.62-1.26]). The interaction *P* value for the relative treatment effect was 0.04, indicating a greater relative reduction in the primary endpoint with bivalirudin compared with heparin in LBR

compared with non-LBR patients. The interaction *P* value for the absolute treatment effect was 0.81, indicating a consistent absolute benefit across the 2 treatment groups for the primary endpoint. Considering the CRUSADE bleeding risk score as a continuous variable, there were no significant interactions between bleeding risk and the treatment effect of bivalirudin vs heparin on the primary endpoint, either on an absolute scale ($P_{\text{absolute interaction}} = 0.38$) or on a relative scale ($P_{\text{relative interaction}} = 0.28$) ([Supplemental Figure 2](#)).

The effects of bivalirudin vs heparin in reducing all-cause death were also at least as strong in LBR patients (32 [1.4%] vs 59 [2.6%], respectively;

TABLE 2 Medication Use in Patients According to the CRUSADE Score and Randomized Anticoagulation Type

	CRUSADE Score <30 (n = 4,581)			CRUSADE Score ≥30 (n = 1,435)		
	Heparin (n = 2,300)	Bivalirudin (n = 2,281)	SMD (95% CI)	Heparin (n = 707)	Bivalirudin (n = 728)	SMD (95% CI)
Medications						
Study medications						
Heparin	2,272 (98.9)	19 (0.8)	-	690 (98.0)	5 (0.7)	-
Total dose during PCI, U	5,600 (4,900-6,900)	-	-	5,250 (4,480-6,300)	-	-
Bivalirudin	26 (1.1)	2,261 (99.2)	-	14 (2.0)	721 (99.3)	-
Post-PCI infusion administered	-	2,237 (98.9)	-	-	716 (99.3)	-
Post-PCI infusion duration, h	-	3.0 (2.1-4.0)	-	-	3.0 (2.2-4.0)	-
Additional bolus of study medications	812 (35.3)	82 (3.6)	-0.87 (-0.94 to -0.81)	242 (34.4)	24 (3.3)	-0.87 (-0.97 to -0.76)
Tirofiban for procedural thrombotic complications	306 (13.3)	245 (10.8)	-0.08 (-0.14 to -0.02)	105 (14.9)	102 (14.1)	-0.02 (-0.13 to 0.08)
Peak activated clotting time, s ^a	268 (238-319)	320 (276-366)	0.26 (0.20 to 0.32)	265 (239-310)	321 (280-365)	0.49 (0.37 to 0.60)
Dual antiplatelet therapy						
Aspirin	2,290 (99.6)	2,267 (99.4)	-0.02 (-0.08 to 0.03)	700 (99.0)	722 (99.2)	0.02 (-0.09 to 0.12)
P2Y ₁₂ inhibitor			-0.01 (-0.06 to 0.05)			-0.04 (-0.14 to 0.06)
Clopidogrel	769 (33.4)	756 (33.1)		264 (37.3)	258 (35.4)	
Ticagrelor	1,531 (66.6)	1,525 (66.9)		443 (62.7)	470 (64.6)	
Invasive procedures						
Arterial access ^b			-0.04 (-0.09 to 0.02)			-0.05 (-0.15 to 0.06)
Transradial	2,153 (93.7)	2,156 (94.6)		627 (89.1)	657 (90.5)	
Transfemoral	145 (6.3)	124 (5.4)		77 (10.9)	69 (9.5)	
Revascularization, any	2,268 (98.6)	2,243 (98.3)	-0.02 (-0.08 to 0.04)	692 (97.9)	710 (97.5)	-0.02 (-0.13 to 0.08)
Coronary arteries treated ^c						
Left main	20 (0.9)	12 (0.5)	-0.04 (-0.10 to 0.02)	12 (1.7)	14 (2.0)	0.02 (-0.09 to 0.12)
Left anterior descending	1,148 (50.6)	1,123 (50.1)	-0.01 (-0.07 to 0.05)	301 (43.5)	334 (47.0)	0.07 (-0.03 to 0.18)
Left circumflex	274 (12.1)	296 (13.2)	0.03 (-0.02 to 0.09)	97 (14.0)	80 (11.3)	-0.08 (-0.19 to 0.02)
Right coronary artery	930 (41.0)	885 (39.5)	-0.03 (-0.09 to 0.03)	316 (45.7)	315 (44.4)	-0.03 (-0.13 to 0.08)
Multivessel intervention	104 (4.6)	72 (3.2)	-0.07 (-0.13 to -0.01)	38 (5.5)	38 (5.4)	-0.01 (-0.11 to 0.10)
Percutaneous coronary intervention	2,260 (98.3)	2,234 (97.9)	-0.02 (-0.08 to 0.03)	689 (97.5)	708 (97.3)	-0.01 (-0.12 to 0.09)
Drug-eluting stent implantation	2,049 (89.1)	2,026 (88.8)	-0.01 (-0.07 to 0.05)	623 (88.1)	641 (88.1)	0.00 (-0.11 to 0.10)
No. of stents	1.3 ± 0.6	1.3 ± 0.5	-0.01 (-0.07 to 0.05)	1.3 ± 0.6	1.3 ± 0.5	-0.07 (-0.18 to 0.04)
Total length of stents, mm	32.7 ± 16.3	32.9 ± 16.0	0.01 (-0.07 to 0.05)	34.7 ± 17.5	33.1 ± 15.8	-0.09 (-0.2 to 0.02)
Balloon angioplasty only	211 (9.2)	208 (9.1)	0.00 (-0.06 to 0.06)	66 (9.3)	67 (9.2)	0.00 (-0.11 to 0.10)
Thrombus aspiration	405 (17.9)	399 (17.9)	0.00 (-0.06 to 0.06)	122 (17.7)	134 (18.9)	0.03 (-0.07 to 0.14)
PCI time intervals						
Symptom onset to wire, h	4.4 (3.0-8.0)	4.5 (2.9-7.8)	-0.02 (-0.08 to 0.05)	4.8 (3.1-8.5)	4.8 (3.0-7.8)	-0.10 (-0.22 to 0.02)
First medical contact to wire, h	1.7 (1.1-3.0)	1.6 (1.1-3.1)	0.02 (-0.05 to 0.08)	1.7 (1.1-3.2)	1.7 (1.2-2.9)	-0.02 (-0.14 to 0.10)
Tertiary hospital door to wire, h	1.1 (0.8-1.7)	1.1 (0.9-1.6)	-0.02 (-0.08 to 0.04)	1.2 (0.9-1.7)	1.2 (0.9-1.8)	-0.02 (-0.14 to 0.09)
Procedure duration, min ^d	29 (20 -41)	29 (20 -42)	-0.03 (-0.09 to 0.03)	30 (20 -43)	29 (20 -43)	-0.03 (-0.15 to 0.08)
PCI TIMI flow, site-assessed						
Pre-PCI			0.05 (-0.01 to 0.11)			0.02 (-0.09 to 0.12)
0	1,716 (76.4)	1,673 (75.2)		527 (77.1)	538 (76.3)	
1	165 (7.4)	155 (7.0)		35 (5.1)	37 (5.3)	
2	151 (6.7)	173 (7.8)		52 (7.6)	54 (7.7)	
3	213 (9.5)	224 (10.1)		70 (10.2)	76 (10.8)	
Post-PCI			0.08 (0.02 to 0.14)			0.19 (0.09 to 0.30)
0	13 (0.6)	10 (0.5)		11 (1.6)	1 (0.1)	
1	1 (0.0)	4 (0.2)		4 (0.6)	0 (0.0)	
2	33 (1.5)	18 (0.8)		11 (1.6)	11 (1.6)	
3	2,204 (97.9)	2,197 (98.6)		656 (96.2)	690 (98.3)	
Staged PCI within 30 days	188 (8.2)	186 (8.2)	0.00 (-0.06 to 0.06)	47 (6.7)	42 (5.8)	-0.04 (-0.14 to 0.07)
Coronary artery bypass graft surgery	8 (0.4)	9 (0.4)	0.01 (-0.05 to 0.07)	3 (0.4)	2 (0.3)	-0.03 (-0.13 to 0.08)
Coronary angiography only	30 (1.3)	37 (1.6)	0.03 (-0.03 to 0.08)	12 (1.7)	16 (2.2)	0.04 (-0.07 to 0.14)
None	2 (0.1)	1 (0.0)	-0.02 (-0.07 to 0.04)	3 (0.4)	2 (0.3)	-0.03 (-0.13 to 0.08)

Values are n (%), median (Q1-Q3) or mean ± SD, unless otherwise indicated. ^aActivated clotting time was measured in 2,685 patients in the heparin group and 2,776 patients in the bivalirudin group. ^bAngiography was not performed in 3 patients in the bivalirudin group and 5 in the heparin group. ^cPer patient; some patients had >1 epicardial coronary artery treated during the index PCI or bypass graft procedure, so the total is greater than 100%. ^dDefined as the time from guiding catheter insertion to its withdrawal.

TIMI = Thrombolysis In Myocardial Infarction; other abbreviations as in Table 1.

TABLE 3 Clinical Outcomes Stratified by the CRUSADE Score

	CRUSADE Score <30 (n = 4,581)	CRUSADE Score ≥30 (n = 1,435)	HR (95% CI)	P Value
Primary endpoint: all-cause death or BARC types 3-5 bleeding	100 (2.2)	124 (8.6)	4.08 (3.14-5.31)	<0.0001
Death from any cause	91 (2.0)	116 (8.1)	4.19 (3.18-5.51)	<0.0001
Death from cardiovascular causes	88 (1.9)	112 (7.8)	4.18 (3.16-5.53)	<0.0001
BARC types 3-5 bleeding	16 (0.3)	13 (0.9)	2.60 (1.25-5.40)	0.01
Access site related	1 (0.02)	0 (0.0)	–	–
Non-access site related	15 (0.3)	13 (0.9)	2.77 (1.32-5.83)	0.007
Reinfarction	26 (0.6)	16 (1.1)	1.97 (1.06-3.68)	0.03
Ischemia-driven target vessel revascularization	21 (0.5)	6 (0.4)	0.91 (0.37-2.26)	0.84
Stroke	22 (0.5)	7 (0.5)	1.02 (0.43-2.38)	0.97
Stent thrombosis	26 (0.6)	18 (1.3)	2.22 (1.21-4.04)	0.01
Acute (<24 h)	12 (0.3)	6 (0.4)	1.60 (0.60-4.25)	0.35
Subacute (1-30 d)	14 (0.3)	12 (0.8)	2.75 (1.27-5.94)	0.01
MACCE ^a	148 (3.2)	130 (9.1)	2.89 (2.28-3.66)	<0.0001
BARC types 2-5 bleeding	92 (2.0)	48 (3.3)	1.68 (1.18-2.37)	0.004
All-cause death or BARC types 2-5 bleeding	174 (3.8)	156 (10.9)	2.96 (2.39-3.68)	<0.0001
Acquired thrombocytopenia ^b	167 (3.7)	53 (3.8)	1.02 (0.75-1.38)	0.92
NACE ^c	157 (3.4)	135 (9.4)	2.83 (2.25-3.56)	<0.0001

Values are n (%) unless otherwise indicated. ^aAll-cause death, myocardial infarction, ischemia-driven target vessel revascularization, or stroke. ^bDefined as nadir platelet count of $<150 \times 10^9/L$ after the index procedure in patients in whom the baseline platelet count was $>150 \times 10^9/L$. ^cMACCE or BARC types 3-5 bleeding.

BARC = Bleeding Academic Research Consortium; MACCE = major adverse cardiac or cerebral events; NACE = net adverse clinical events; other abbreviations as in [Table 1](#).

difference: -1.2% [95% CI: -2.0% to -0.4%]; HR: 0.54 [95% CI: 0.35-0.84]) as in non-LBR patients (57 [7.8%] vs 59 [8.3%], respectively; difference: -0.5% [95% CI: -3.3% to 2.3%]; HR: 0.94 [95% CI: 0.66-1.36]) patients ($P_{\text{absolute interaction}} = 0.67$; $P_{\text{relative interaction}} = 0.06$). The effects of bivalirudin vs heparin in reducing BARC types 3 to 5 bleeding were consistent in LBR patients (2 [0.1%] vs 14 [0.6%], respectively; difference: -0.5% [95% CI: -0.9% to -0.2%]; HR 0.14 [95% CI: 0.03-0.63]) and non-LBR patients (3 [0.4%] vs 10 [1.4%], respectively; difference: -1.0% [95% CI: -2.0% to -0.02%]; HR: 0.29 [95% CI: 0.08-1.05]) ($P_{\text{absolute interaction}} = 0.37$; $P_{\text{relative interaction}} = 0.48$), owing to reductions in both risk strata in nonaccess site bleeding with bivalirudin. The effects of bivalirudin vs heparin treatment were consistent in LBR and non-LBR patients for all other secondary outcomes examined ([Table 4](#)).

SENSITIVITY ANALYSIS. As with the CRUSADE risk score, patients categorized as non-LBR by the PRECISE-DAPT score and the ARC-HBR criteria had substantially higher rates of the 30-day primary composite outcome compared with LBR patients ([Supplemental Table 3](#)). Compared with heparin monotherapy, bivalirudin reduced the relative risk of the primary endpoint to a similar degree in LBR and non-LBR patients as stratified by both the PRECISE-DAPT score and the ARC-HBR criteria

([Supplemental Table 4](#)). The absolute risk of the primary endpoint was reduced to a similar degree in LBR and non-LBR patients when stratified by the PRECISE-DAPT score, but to a greater degree in non-LBR patients when stratified by the ARC-HBR criteria ([Supplemental Table 4](#)).

DISCUSSION

The principal findings from this prespecified analysis from BRIGHT-4 in which outcomes were examined according to baseline bleeding risk are as follows ([Central Illustration](#)). First, the widely used CRUSADE bleeding risk instrument not only effectively stratified STEMI patients at baseline who were at high risk for major and minor bleeding within 30 days after primary PCI, but also identified patients at increased risk for early reinfarction and stent thrombosis, as well as cardiovascular and all-cause death. Second, treatment with bivalirudin compared with heparin had robust absolute and relative effects in reducing the 30-day primary composite endpoint of all-cause death or major bleeding and all-cause death and bleeding separately in LBR as well as in non-LBR patients. Third, despite the higher 30-day rates of adverse outcomes in non-LBR compared with LBR patients, given the 3:1 prevalence of LBR to non-LBR patients and the relative risks observed, more deaths and major bleeding were prevented by bivalirudin

TABLE 4 Clinical Outcomes at 30 Days Stratified by CRUSADE Score and Anticoagulation Type

	CRUSADE Score <30 (n = 4,581)				CRUSADE Score ≥30 (n = 1,435)				<i>P</i> _{absolute interaction}	<i>P</i> _{relative interaction}
	Heparin (n = 2,300)	Bivalirudin (n = 2,281)	Absolute Difference, % (95% CI)	HR (95% CI)	Heparin (n = 707)	Bivalirudin (n = 728)	Absolute Difference, % (95% CI)	HR (95% CI)		
Primary endpoint: all-cause death or BARC types 3-5 bleeding	67 (2.9)	33 (1.4)	−1.5 (−2.3 to −0.6)	0.49 (0.32-0.75)	65 (9.2)	59 (8.1)	−1.1 (−4.0 to 1.8)	0.88 (0.62-1.26)	0.81	0.04
Death from any cause	59 (2.6)	32 (1.4)	−1.2 (−2.0 to −0.4)	0.54 (0.35-0.84)	59 (8.3)	57 (7.8)	−0.5 (−3.3 to 2.3)	0.94 (0.66-1.36)	0.67	0.06
Death from cardiovascular causes	56 (2.4)	32 (1.4)	−1.0 (−1.8 to −0.2)	0.57 (0.37-0.88)	57 (8.1)	55 (7.6)	−0.5 (−3.3 to 2.3)	0.94 (0.65-1.36)	0.72	0.09
BARC types 3-5 bleeding	14 (0.6)	2 (0.1)	−0.5 (−0.9 to −0.2)	0.14 (0.03-0.63)	10 (1.4)	3 (0.4)	−1.0 (−2.0 to −0.02)	0.29 (0.08-1.05)	0.37	0.48
Access site related	1 (0.04)	0 (0.0)	−0.04 (−0.1 to 0.04)	− (0.04-0.69)	0 (0.0)	0 (0.0)	− (−2.0 to −0.02)	− (0.08-1.06)	0.82	0.99
Non-access site related	13 (0.6)	2 (0.1)	−0.5 (−0.8 to −0.1)	0.16 (0.04-0.69)	10 (1.4)	3 (0.4)	−1.0 (−2.0 to −0.02)	0.29 (0.08-1.06)	0.32	0.53
Reinfarction	14 (0.6)	12 (0.5)	−0.1 (−0.5 to 0.4)	0.86 (0.40-1.87)	11 (1.6)	5 (0.7)	−0.9 (−2.0 to 0.2)	0.44 (0.15-1.27)	0.19	0.31
Ischemia-driven TVR	15 (0.7)	6 (0.3)	−0.4 (−0.8 to 0.00)	0.40 (0.16-1.04)	3 (0.4)	3 (0.4)	−0.01 (−0.7 to 0.7)	0.97 (0.20-4.81)	0.34	0.35
Stroke	10 (0.4)	12 (0.5)	0.1 (−0.3 to 0.5)	1.21 (0.52-2.80)	4 (0.6)	3 (0.4)	−0.2 (−0.9 to 0.6)	0.73 (0.16-3.26)	0.56	0.56
Stent thrombosis	20 (0.9)	6 (0.3)	−0.6 (−1.0 to −0.2)	0.30 (0.12-0.75)	13 (1.8)	5 (0.7)	−1.2 (−2.3 to 0.0)	0.37 (0.13-1.04)	0.39	0.77
Acute (<24 h)	10 (0.4)	2 (0.1)	−0.3 (−0.6 to −0.1)	0.20 (0.04-0.92)	4 (0.6)	2 (0.3)	−0.3 (−1.0 to 0.4)	0.49 (0.09-2.65)	0.88	0.45
Subacute (1-30 d)	10 (0.4)	4 (0.2)	−0.3 (−0.6 to 0.1)	0.40 (0.13-1.28)	9 (1.3)	3 (0.4)	−0.9 (−1.8 to 0.1)	0.32 (0.09-1.19)	0.24	0.80
MACCE ^a	89 (3.9)	59 (2.6)	−1.3 (−2.3 to −0.3)	0.66 (0.48-0.92)	66 (9.3)	64 (8.8)	−0.5 (−3.5 to 2.4)	0.95 (0.67-1.34)	0.64	0.14
BARC types 2-5 bleeding	53 (2.3)	39 (1.7)	−0.6 (−1.4 to 0.2)	0.74 (0.49-1.12)	24 (3.4)	24 (3.3)	−0.1 (−2.0 to 1.8)	0.98 (0.55-1.72)	0.63	0.44
All-cause death or BARC types 2-5 bleeding	105 (4.6)	69 (3.0)	−1.5 (−2.6 to −0.4)	0.66 (0.49-0.89)	78 (11.0)	78 (10.7)	−0.3 (−3.5 to 2.9)	0.98 (0.72-1.34)	0.48	0.07
Acquired thrombocytopenia ^b	94 (4.2)	73 (3.3)	−0.9 (−2.0 to 0.2)	0.78 (0.58-1.06)	29 (4.2)	24 (3.4)	−0.8 (−2.8 to 1.2)	0.80 (0.47-1.38)	0.95	0.94
NACE ^c	97 (4.2)	60 (2.6)	−1.6 (−2.6 to −0.5)	0.62 (0.45-0.85)	70 (9.9)	65 (8.9)	−1.0 (−4.0 to 2.1)	0.90 (0.65-1.27)	0.71	0.11

Values are n (%) unless otherwise indicated. Event rates are number of events (Kaplan-Meier estimated percentages). ^aAll-cause death, myocardial infarction, ischemia-driven target vessel revascularization, or stroke. ^bDefined as nadir platelet count of <150×10⁹/L after the index procedure in patients in whom the baseline platelet count was >150×10⁹/L. ^cMACCE or BARC types 3 to 5 bleeding. TVR = target vessel revascularization; other abbreviations as in [Tables 1 and 3](#).

treatment in the LBR cohort than in the non-LBR cohort.

Previous studies have consistently demonstrated that major bleeding is independently associated with increased mortality and ischemic complications after primary PCI in STEMI,²⁴ underscoring the importance of selecting the optimal procedural anticoagulation regimen to balance thrombotic and bleeding risks. In early primary PCI trials in STEMI, the short half-life of bivalirudin led to increased rates of acute (<24 h) stent thrombosis when its use was restricted to the catheterization laboratory compared with heparin plus the routine use of glycoprotein IIb/IIIa inhibitors.³⁻⁵ A meta-analysis⁹ of 4 randomized trials^{4,6-8} subsequently showed that adding a high-dose post-PCI bivalirudin infusion for 2 to 4 hours

prevented this excess thrombotic risk, and that with this regimen, bivalirudin reduced major bleeding and cardiac death compared with heparin monotherapy in STEMI. Yet, an adequately powered trial of these 2 favored regimens had not been performed, and whether the benefits of bivalirudin would extend to patients undergoing primary PCI with radial artery access, a strategy that virtually eliminates access site bleeding and improves survival,^{25,26} remained uncertain. The subsequently performed BRIGHT-4 trial in which 6,016 patients with STEMI undergoing primary PCI with radial artery access in >93% of cases were randomized to bivalirudin plus a post-PCI high-dose infusion for 2 to 4 hours vs heparin monotherapy answered this question, demonstrating reductions in major and minor bleeding with

bivalirudin owing to less non-access site-related bleeding,¹⁰ the occurrence of which has been more strongly associated with subsequent mortality than access site-related bleeding.^{24,27,28} The extended post-PCI bivalirudin infusion, coupled with its inherent antiplatelet effects²⁹ also led to a reduction in stent thrombosis with bivalirudin compared with heparin monotherapy,¹⁰ which paradoxically activates platelets and increases thrombotic risk if a glycoprotein IIb/IIIa inhibitor is not routinely used.³⁰ The net effects of reduced major bleeding and stent thrombosis resulted in decreased 30-day mortality with procedural anticoagulation with bivalirudin compared with heparin monotherapy during primary PCI in BRIGHT-4 (3.0% vs 3.9%; HR: 0.75 [95% CI: 0.57-0.99]; $P = 0.04$).¹⁰ However, given the strong relationship between reduced bleeding and mortality in STEMI, whether bivalirudin would have similar effects in patients at lower-risk as well as in those at higher-risk for bleeding was unknown, prompting the present analysis.

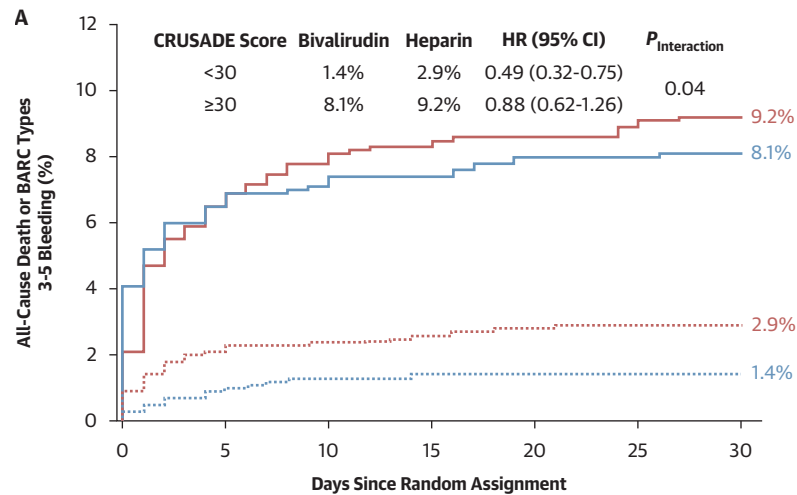
While several bleeding risk scores have been developed and shown to provide reliable estimates of long-term post-discharge bleeding risk in patients with acute coronary syndromes and after PCI,^{19,20,31-34} the widely used CRUSADE score was developed and validated specifically for the prediction of in-hospital major bleeding risk. We therefore prespecified use of the CRUSADE score for the planned bleeding risk substudy. The present analysis confirms that a CRUSADE score ≥ 30 has good discrimination for predicting 30-day major bleeding, all-cause death and the composite of all-cause death or major bleeding after primary PCI in STEMI in the Chinese population enrolled in BRIGHT-4, despite concerns that ethnic and regional variations in baseline characteristics may influence its predictive accuracy.³⁵ In addition, BRIGHT-4 patients who were at non-LBR by the CRUSADE tool had higher rates of stent thrombosis and reinfarction, MACCE and NACE compared with LBR patients.

Importantly, the relative treatment effects of bivalirudin in reducing the 30-day rates of the primary endpoint, major bleeding, all-cause death, as well as minor bleeding, stent thrombosis, cardiovascular death, and NACE compared with heparin in BRIGHT-4¹⁰ were robust in CRUSADE-stratified LBR patients as well as non-LBR patients. Notably, the HR favoring bivalirudin for reduction in the 30-day primary endpoint was lower (ie, the relative treatment effect was greater) in LBR than in non-LBR patients (HR: 0.49 [95% CI: 0.32-0.75] vs HR: 0.88 [95% CI: 0.62-1.26]; $P_{\text{relative interaction}} = 0.04$). Similarly, the HR favoring bivalirudin for reduction in the 30-day risk

of all-cause death was 0.54 (95% CI: 0.35-0.84) in LBR patients compared with 0.94 (95% CI: 0.66-1.36) in non-LBR patients ($P_{\text{relative interaction}} = 0.06$). As a result, the absolute benefits for these endpoints with bivalirudin compared with heparin were similar in LBR and non-LBR patients (absolute reductions with bivalirudin in 1.5% vs 1.1% of patients, respectively; $P_{\text{absolute interaction}} = 0.81$). Spline analysis demonstrated consistent relative and absolute benefits of bivalirudin compared with heparin across the entire spectrum of baseline bleeding risk as defined by the CRUSADE score. Most importantly, given the fact that the 76% of patients were at low risk for bleeding, 28 more primary outcome events were prevented with bivalirudin compared with heparin treatment in the LBR than in the non-LBR cohort (34 vs 6 fewer deaths or major bleeding, respectively), and 25 more deaths were prevented with bivalirudin compared with heparin treatment in the LBR than in the non-LBR cohort. Because BRIGHT-4 had few enrollment restrictions, these data emphasize that in addition to non-LBR patients, treatment of LBR patients with bivalirudin rather than heparin would be expected to prevent death and bleeding in the largest number of patients.

Regarding bleeding, the relative and absolute treatment effects of bivalirudin vs heparin were consistent across the 2 CRUSADE bleeding risk strata ($P_{\text{relative interaction}} = 0.48$; $P_{\text{absolute interaction}} = 0.37$). However, because there were 3 times as many LBR as non-LBR patients, 12 LBR vs 7 non-LBR patients had a major bleeding prevented with treatment with bivalirudin rather than heparin. Of note, only a single major bleeding arose from the access site of any patient in the entire BRIGHT-4 trial, owing to the near universal use of radial artery access; thus, the benefit of bivalirudin in reducing major bleeding was restricted to non-access-site-related bleeding. An even greater absolute major bleeding benefit would likely have occurred had femoral access been more frequently used.

To our knowledge the relationship between bleeding risk and the utility of bivalirudin vs heparin anticoagulation during primary PCI for STEMI has been examined in only 1 previous study: the first BRIGHT trial (NCT01696110).⁷ In BRIGHT, 2,194 STEMI patients were randomized 1:1:1 to bivalirudin with a 2 to 4-hour post-PCI high-dose infusion ($n = 735$) vs heparin plus tirofiban ($n = 730$) vs heparin monotherapy ($n = 729$). The relative reduction in BARC types 2-5 bleeding with bivalirudin compared with heparin $\pm$ tirofiban were similar in patients with CRUSADE scores <30 and ≥ 30 , although MACCE and NACE were reduced to a greater relative extent with

FIGURE 3 30-Day Outcomes in Patients Treated With Bivalirudin Plus a Post-PCI High-Dose Infusion vs Heparin Monotherapy, Stratified by the CRUSADE Bleeding Risk Score**Number at risk**

CRUSADE score <30

2,300	2,252	2,244	2,241	2,236	2,233	2,233
-------	-------	-------	-------	-------	-------	-------

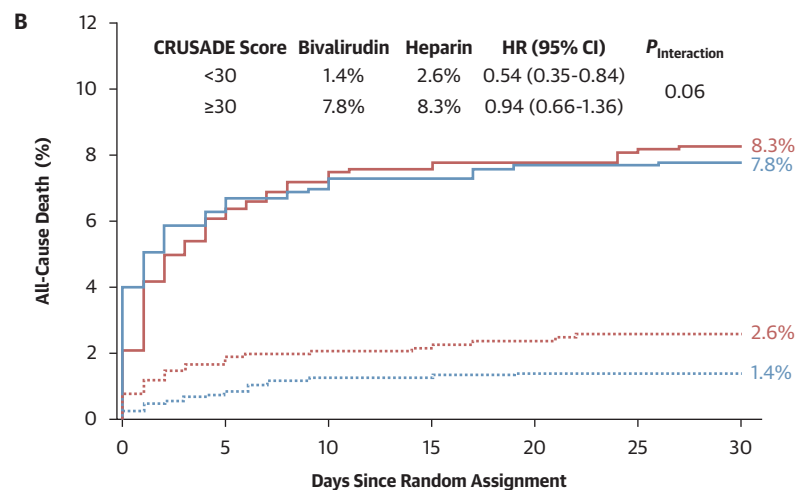
2,281	2,261	2,251	2,250	2,249	2,248	2,248
-------	-------	-------	-------	-------	-------	-------

CRUSADE score ≥ 30

707	661	652	648	646	644	642
-----	-----	-----	-----	-----	-----	-----

728	681	676	674	670	670	669
-----	-----	-----	-----	-----	-----	-----

..... Bivalirudin-CRUSADE <30 — Bivalirudin-CRUSADE ≥ 30
 Heparin-CRUSADE <30 — Heparin-CRUSADE ≥ 30

**Number at risk**

CRUSADE score <30

2,300	2,259	2,252	2,249	2,244	2,241	2,241
-------	-------	-------	-------	-------	-------	-------

2,281	2,252	2,250	2,250	2,250	2,249	2,249
-------	-------	-------	-------	-------	-------	-------

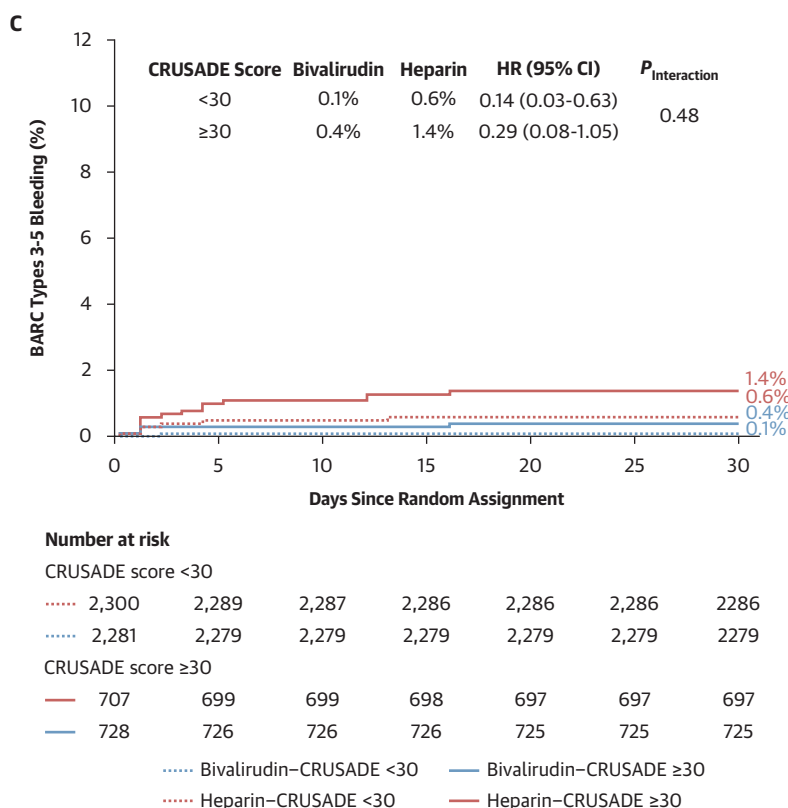
CRUSADE score ≥ 30

707	663	655	652	652	649	648
-----	-----	-----	-----	-----	-----	-----

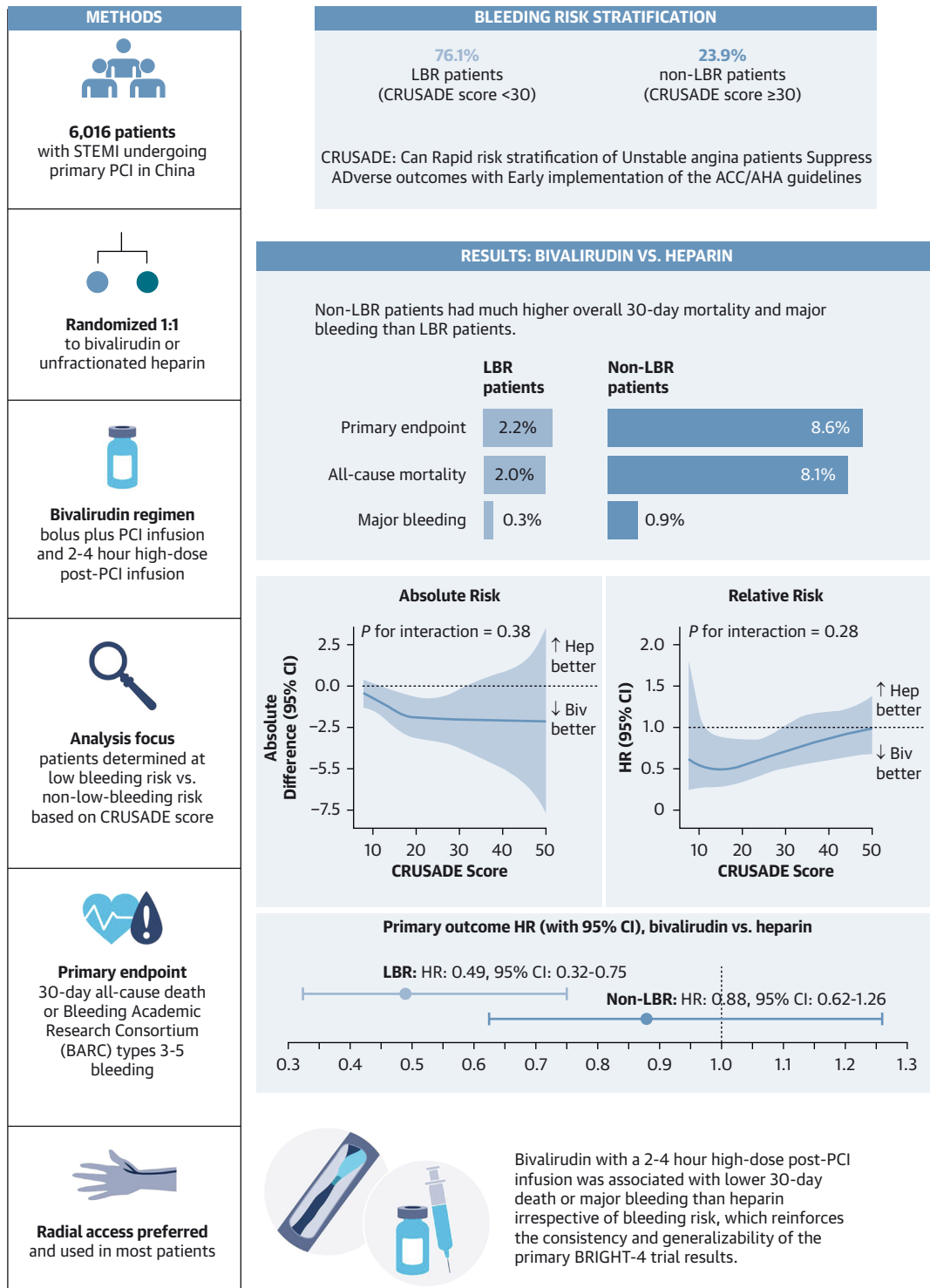
728	681	676	675	672	672	671
-----	-----	-----	-----	-----	-----	-----

..... Bivalirudin-CRUSADE <30 — Bivalirudin-CRUSADE ≥ 30
 Heparin-CRUSADE <30 — Heparin-CRUSADE ≥ 30

(A) The primary outcome (composite of all-cause death or BARC types 3-5 bleeding); (B) all-cause death; (C) BARC types 3 to 5 bleeding.



STUDY LIMITATIONS. First, randomization in BRIGHT-4 was not stratified by bleeding risk, and although baseline clinical variables were well

CENTRAL ILLUSTRATION Prespecified Analysis From the BRIGHT-4 Trial According to Bleeding Risk

JACC & ESC 2026: Qiu M, Duan Y, Feng X, et al. Bivalirudin Versus Heparin in Low and Non-Low Bleeding Risk Patients Undergoing Primary PCI for STEMI: The BRIGHT-4 trial

Qiu M, et al. JACC. 2026;■(■):■-■.

A total of 6,016 patients with STEMI undergoing primary PCI with radial artery access were randomized to anticoagulation with bivalirudin plus a high-dose post-PCI infusion for 2 to 4 hours vs heparin monotherapy. The primary endpoint was the 30-day composite occurrence of all-cause death or BARC types 3 to 5 bleeding. Bivalirudin reduced the primary endpoint in patients at low bleeding risk as well as in patients at high bleeding risk according to the baseline CRUSADE score. LBR = low bleeding risk; STEMI = ST-segment elevation myocardial infarction.

matched between the 2 treatment groups in each bleeding risk stratum, small differences in unmeasured confounders cannot be excluded. Second, the present study focused on 30-day outcomes, the time course most likely to be affected by procedural anticoagulation strategies. However, the adverse effects of bleeding and recurrent ischemic events may be prolonged, and further studies with longer follow-up are warranted. Third, although generalizability may be a concern because only patients from China were enrolled, the BRIGHT-4 findings have been replicated in previous U.S. and EU trials.⁹ Fourth, subtle differences in absolute and relative treatment effects with bivalirudin treatment were present in LBR vs non-LBR patients as categorized by the prespecified CRUSADE risk score vs in post hoc analyses using the PRECISE-DAPT score and ARC-HBR criteria. In addition to the varying components that compose the 3 scores, these differences may reflect the fact that the CRUSADE risk score was developed to predict in-hospital bleeding whereas the PRECISE-DAPT and ARC-HBR scores were designed for long-term bleeding risk. Nonetheless, the 30-day primary composite outcome as well as all-cause death and major bleeding were reduced with bivalirudin across the spectrum of bleeding risk with all 3 risk scores. Finally, the effects of bivalirudin vs heparin were not powered in the separate non-LBR and LBR cohorts; their proper interpretation therefore relies on examining consistency of treatment effects in each subgroup. The approximate 3:1 distribution of LBR to non-LBR patients reduces power for interaction testing. Conversely, interaction testing was not adjusted for multiplicity and therefore type I error for some of these findings cannot be excluded. The results from the present study should therefore be considered as hypothesis generating, warranting further investigation in additional prospective studies.

CONCLUSIONS

In this prespecified analysis from the BRIGHT-4 trial, patients with STEMI undergoing primary PCI with radial artery access who at baseline were not at LBR according to a CRUSADE score ≥ 30 had markedly

higher risks of mortality, bleeding, and adverse ischemic outcomes after primary PCI compared with LBR patients. Patients randomized to procedural anticoagulation with bivalirudin plus a post-PCI high-dose infusion for 2 to 4 hours had reduced 30-day rates of the primary composite outcome of all-cause death or BARC types 3 to 5 bleeding, as well as reductions in major and minor bleeding, stent thrombosis, cardiovascular death, and all-cause death compared with heparin monotherapy, regardless of bleeding risk. These findings reinforce the consistency and generalizability of the primary BRIGHT-4 trial results across the spectrum of baseline bleeding risk in patients with STEMI undergoing primary PCI.

DATA AVAILABILITY The datasets used and analyzed during this study may be made available from the corresponding author on reasonable request.

ACKNOWLEDGMENTS The principal investigators thank the participating physicians and research coordinators from each site.

FUNDING SUPPORT AND AUTHOR DISCLOSURES

This work was supported by grants from National Key Technology Research and Development Program of China (2024ZD0537904), Chinese Society of Cardiology Foundation (CSCF2019A01), and a research grant from Jiangsu Hengrui Pharmaceuticals. Dr Stone has received speaker honoraria from Boehringer Ingelheim; has served as a consultant for Millennia Biopharma, Apollo Therapeutics, Cardiac Success, Occlutech, Oxitope, Elixir, Impulse Dynamics, Asceneuron, Myochron, Vesalio, HeartFlow, Colibri, Bioventrix, Abbott, Remote Cardiac Enablement, Valfix, Zoll, Shockwave, Adona Medical, HighLife, Elucid Bio, Aria, Alleviant, FBR Medical, MedHub, Biotyx Medical, and Ablative Solutions; has equity and options from Cardiac Success, Ancora, Cagent, Applied Therapeutics, Biostar family of funds, SpectraWave, Orchestra Biomed, Aria, Valfix, Xenter, and Vascantis; and his employer, Mount Sinai Hospital, has received research grants from Shockwave, Biosense-Webster, Bioventrix, Abbott, Abiomed, Cardiovascular Systems Inc, Phillips, Pulnovo, V-wave, and PCORI (via Weill Cornell Medical Center). All other authors have reported that they have no relationships relevant to the contents of this paper to disclose.

ADDRESS FOR CORRESPONDENCE: Dr Yaling Han, State Key Laboratory of Frigid Zone Cardiovascular Disease, Department of Cardiology, General Hospital of Northern Theater Command, Shenyang 110016, China. E-mail: hanyaling@163.net.

REFERENCES

1. O'Gara PT, Kushner FG, Ascheim DD, et al. 2013 ACCF/AHA guideline for the management of ST-elevation myocardial infarction: a report of the American College of Cardiology Foundation/American Heart Association Task Force on Practice Guidelines. *J Am Coll Cardiol*. 2013;61:e78-e140.
2. Ibanez B, James S, Agewall S, et al. 2017 ESC guidelines for the management of acute myocardial infarction in patients presenting with ST-segment elevation: the Task Force for the Management of Acute Myocardial Infarction in Patients Presenting With ST-Segment Elevation of the European Society of Cardiology (ESC). *Eur Heart J*. 2018;39:119-177.
3. Stone GW, Witzenbichler B, Guagliumi G, et al. Bivalirudin during primary PCI in acute myocardial infarction. *N Engl J Med*. 2008;358:2218-2230.

4. Steg PG, van 't Hof A, Hamm CW, et al. Bivalirudin started during emergency transport for primary PCI. *N Engl J Med*. 2013;369:2207-2217.
5. Shahzad A, Kemp I, Mars C, et al. Unfractionated heparin versus bivalirudin in primary percutaneous coronary intervention (HEAT-PPCI): an open-label, single centre, randomised controlled trial. *Lancet*. 2014;384:1849-1858.
6. Valgimigli M, Frigoli E, Leonardi S, et al. Bivalirudin or unfractionated heparin in acute coronary syndromes. *N Engl J Med*. 2015;373:997-1009.
7. Han Y, Guo J, Zheng Y, et al. Bivalirudin vs heparin with or without tirofiban during primary percutaneous coronary intervention in acute myocardial infarction: the BRIGHT randomized clinical trial. *JAMA*. 2015;313:1336-1346.
8. Erlinge D, Omerovic E, Frobert O, et al. Bivalirudin versus heparin monotherapy in myocardial infarction. *N Engl J Med*. 2017;377:1132-1142.
9. Stone GW, Valgimigli M, Erlinge D, et al. Bivalirudin vs heparin anticoagulation in STEMI: confirmation of the BRIGHT-4 results. *J Am Coll Cardiol*. 2024;84:1512-1524.
10. Li Y, Liang Z, Qin L, et al. Bivalirudin plus a high-dose infusion versus heparin monotherapy in patients with ST-segment elevation myocardial infarction undergoing primary percutaneous coronary intervention: a randomised trial. *Lancet*. 2022;400:1847-1857.
11. Rao SV, O'Donoghue ML, Ruel M, et al. 2025 ACC/AHA/ACEP/NAEMSP/SCAI guideline for the management of patients with acute coronary syndromes: a report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. *J Am Coll Cardiol*. 2025;85:2135-2237.
12. Kesti H, Mattila K, Jaakkola S, et al. Impact of high bleeding risk and associated risk factors on major adverse cardiovascular or cerebrovascular events in primary percutaneous coronary intervention treated ST-elevation myocardial infarction. *Int J Cardiol*. 2025;422:132986.
13. Nicolas J, Beerkens F, Cao D, et al. Performance of the academic research consortium high-bleeding risk criteria in patients undergoing PCI for acute myocardial infarction. *J Thromb Thrombolysis*. 2022;53:20-29.
14. Liang Z, Li Y, Stone GW, Han Y. Rationale and design of a randomized controlled trial of Bivalirudin with a prolonged high-dose infusion versus heparin monotherapy during primary percutaneous coronary intervention in patients with acute ST-Segment elevation myocardial infarction: the BRIGHT-4 Trial. *Cardiol Discov*. 2022;2:226-230.
15. Subherwal S, Bach RG, Chen AY, et al. Baseline risk of major bleeding in non-ST-segment-elevation myocardial infarction: the CRUSADE (Can Rapid Risk Stratification of Unstable Angina Patients Suppress Adverse Outcomes With Early Implementation of the ACC/AHA Guidelines) bleeding score. *Circulation*. 2009;119:1873-1882.
16. Abu-Assi E, Gracia-Acuna JM, Ferreira-Gonzalez I, Pena-Gil C, Gayoso-Diz P, Gonzalez-Juanatey JR. Evaluating the performance of the Can Rapid Risk Stratification of Unstable Angina Patients Suppress Adverse Outcomes With Early Implementation of the ACC/AHA Guidelines (CRUSADE) bleeding score in a contemporary Spanish cohort of patients with non-ST-segment elevation acute myocardial infarction. *Circulation*. 2010;121:2419-2426.
17. Park KH, Jeong MH, Ahn Y, et al. The impact of vascular access for in-hospital major bleeding in patients with acute coronary syndrome at moderate to very high bleeding risk. *J Korean Med Sci*. 2013;28:1307-1315.
18. Liu R, Zheng W, Zhao G, et al. Predictive validity of CRUSADE, ACTION and ACUITY-HORIZONS bleeding risk scores in chinese patients with ST-segment elevation myocardial infarction. *Circ J*. 2018;82:791-797.
19. Costa F, van Klaveren D, James S, et al. Derivation and validation of the predicting bleeding complications in patients undergoing stent implantation and subsequent dual anti-platelet therapy (PRECISE-DAPT) score: a pooled analysis of individual-patient datasets from clinical trials. *Lancet*. 2017;389:1025-1034.
20. Urban P, Mehran R, Collieran R, et al. Defining high bleeding risk in patients undergoing percutaneous coronary intervention. *Circulation*. 2019;140:240-261.
21. Cutlip DE, Windecker S, Mehran R, et al. Clinical endpoints in coronary stent trials: a case for standardized definitions. *Circulation*. 2007;115:2344-2351.
22. Wang TY, Ou FS, Roe MT, et al. Incidence and prognostic significance of thrombocytopenia developed during acute coronary syndrome in contemporary clinical practice. *Circulation*. 2009;119:2454-2462.
23. Mehran R, Rao SV, Bhatt DL, et al. Standardized bleeding definitions for cardiovascular clinical trials: a consensus report from the Bleeding Academic Research Consortium. *Circulation*. 2011;123:2736-2747.
24. Kwok CS, Khan MA, Rao SV, et al. Access and nonaccess site bleeding after percutaneous coronary intervention and risk of subsequent mortality and major adverse cardiovascular events: systematic review and meta-analysis. *Circ Cardiovasc Interv*. 2015;8:e001645.
25. Valgimigli M, Frigoli E, Leonardi S, et al. Radial versus femoral access and bivalirudin versus unfractionated heparin in invasively managed patients with acute coronary syndrome (MATRIX): final 1-year results of a multicentre, randomised controlled trial. *Lancet*. 2018;392:835-848.
26. Mason PJ, Shah B, Tamis-Holland JE, et al. An update on radial artery access and best practices for transradial coronary angiography and intervention in acute coronary syndrome: a scientific statement from the American Heart Association. *Circ Cardiovasc Interv*. 2018;11:e000035.
27. Verheugt FW, Steinhubl SR, Hamon M, et al. Incidence, prognostic impact, and influence of antithrombotic therapy on access and nonaccess site bleeding in percutaneous coronary intervention. *JACC Cardiovasc Interv*. 2011;4:191-197.
28. Vranckx P, Campo G, Anselmi M, et al. Does the site of bleeding matter? A stratified analysis on location of TIMI-graded bleedings and their impact on 12-month outcome in patients with ST-segment elevation myocardial infarction. *Euro-Intervention*. 2012;8:71-78.
29. Anand SX, Kim MC, Kamran M, et al. Comparison of platelet function and morphology in patients undergoing percutaneous coronary intervention receiving bivalirudin versus unfractionated heparin versus clopidogrel pretreatment and bivalirudin. *Am J Cardiol*. 2007;100:417-424.
30. Gao C, Boylan B, Fang J, Wilcox DA, Newman DK, Newman PJ. Heparin promotes platelet responsiveness by potentiating $\alpha IIb\beta 3$ -mediated outside-in signaling. *Blood*. 2011;117:4946-4952.
31. Mehran R, Pocock SJ, Nikolsky E, et al. A risk score to predict bleeding in patients with acute coronary syndromes. *J Am Coll Cardiol*. 2010;55:2556-2566.
32. Mehran R, Pocock S, Nikolsky E, et al. Impact of bleeding on mortality after percutaneous coronary intervention results from a patient-level pooled analysis of the REPLACE-2 (Randomized Evaluation of PCI Linking Angiomax to Reduced Clinical Events), ACUITY (Acute Catheterization and Urgent Intervention Triage Strategy), and HORIZONS-AMI (Harmonizing Outcomes With Revascularization and Stents in Acute Myocardial Infarction) trials. *JACC Cardiovasc Interv*. 2011;4:654-664.
33. Baber U, Mehran R, Giustino G, et al. Coronary thrombosis and major bleeding after PCI with drug-eluting stents: risk scores from PARIS. *J Am Coll Cardiol*. 2016;67:2224-2234.
34. Kawashima H, Gao C, Takahashi K, et al. Comparative assessment of predictive performance of PRECISE-DAPT, CRUSADE, and ACUITY scores in risk-stratifying 30-day bleeding events. *Thromb Haemost*. 2020;120:1087-1095.
35. Wang TKM, Mehta OH, Liao YB, Wang MTM, Stewart R, White H. Meta-analysis of bleeding scores performance for acute coronary syndrome. *Heart Lung Circ*. 2020;29:1749-1757.
36. Amon J, Wong GC, Lee T, et al. Incidence and predictors of adverse events among initially stable ST-elevation myocardial infarction patients following primary percutaneous coronary intervention. *J Am Heart Assoc*. 2022;11:e025572.
37. Tobe A, Sawano M, Kohsaka S, et al. Ischemic and bleeding outcomes in patients who underwent percutaneous coronary intervention with chronic kidney disease or dialysis (from a Japanese nationwide registry). *Am J Cardiol*. 2023;195:37-44.

38. Gharacholou SM, Lopes RD, Alexander KP, et al. Age and outcomes in ST-segment elevation myocardial infarction treated with primary percutaneous coronary intervention: findings from the APEX-AMI trial. *Arch Intern Med*. 2011;171:559–567.

39. Damluji AA, Nanna MG, Mason P, et al. Coronary artery revascularization in the older adult population: a scientific statement from the

American Heart Association. *Circulation*. 2025;152:e494–e525.

40. Stone GW, Clayton T, Deliargyris EN, Prats J, Mehran R, Pocock SJ. Reduction in cardiac mortality with bivalirudin in patients with and without major bleeding: the HORIZONS-AMI trial (Harmonizing Outcomes With Revascularization and Stents in Acute Myocardial Infarction). *J Am Coll Cardiol*. 2014;63:15–20.

KEY WORDS bivalirudin, bleeding, CRUSADE score, heparin, mortality, percutaneous coronary intervention

APPENDIX For supplemental tables and figures and the study protocol, please see the online version of this paper.

Bivalirudin Versus Heparin in Low and Non-Low Bleeding Risk Patients Undergoing Primary PCI for STEMI: The BRIGHT-4 Trial

Miaohan Qiu, Yixuan Duan, Xiaojian Feng, Kui Chen, Dingbao Zhang, Yuncheng Zou, Xiaohui Zheng, Gang Zhao, Nailiang Tian, Zeqi Zheng, Xiaoping Peng, Qing Yang, Zhenyang Liang, Yi Li, Yaling Han, Gregg W. Stone

In the BRIGHT-4 trial, among 6,016 patients with ST-segment elevation myocardial infarction undergoing primary percutaneous coronary intervention (PCI) via radial access, procedural anticoagulation with bivalirudin plus a 2 to 4-hour high-dose post-PCI infusion reduced 30-day death or major bleeding compared with heparin monotherapy. The benefit of bivalirudin in reducing both the composite outcome of mortality or major bleeding as well as all-cause mortality was present across the full spectrum of baseline bleeding risk, including patients at low bleeding risk.

0000